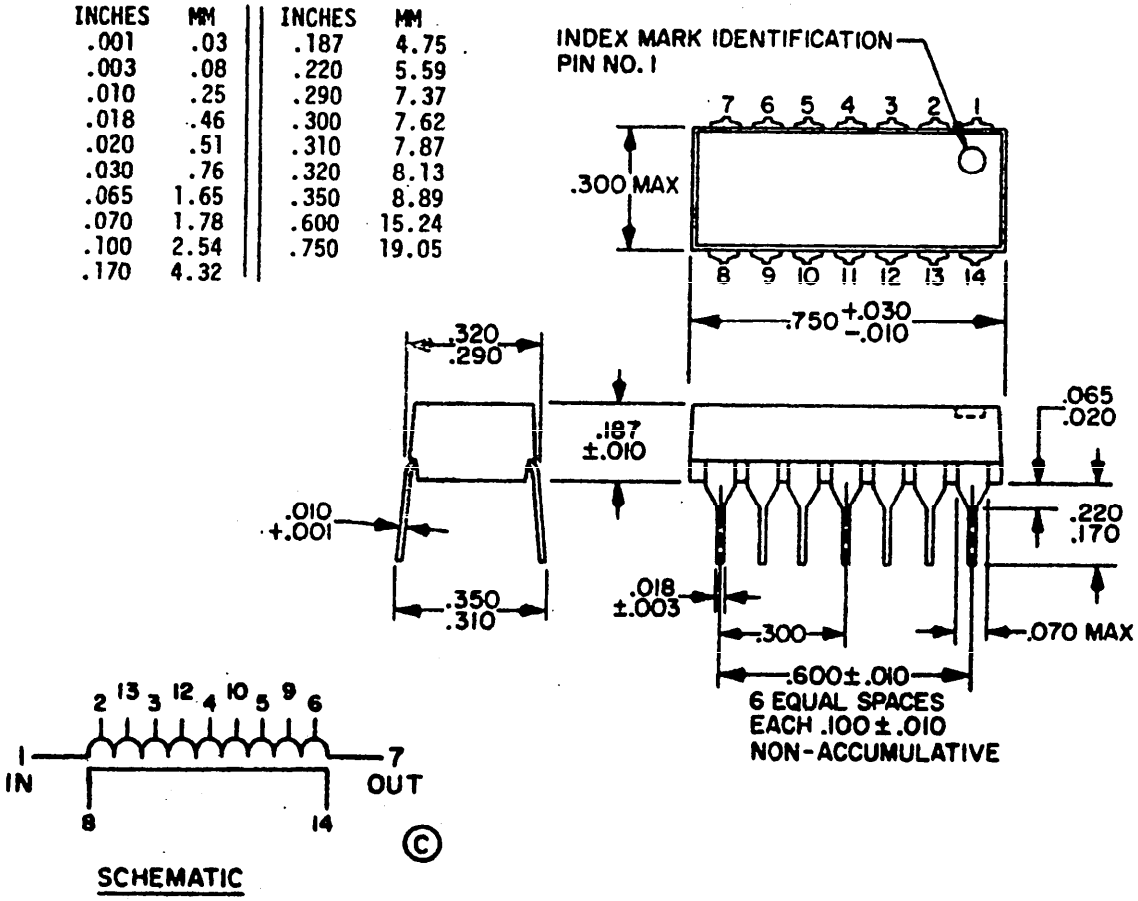
MILITARY SPECIFICATION SHEET

DELAY LINES, PULSE, ELECTROMAGNETIC, LUMPED CONSTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the delay lines described herein shall consist of this document and the latest issue of Specification MIL-D-23859.

INCHES	MM	INCHES	MM
.001	.03	.187	4.75
.003	.08	.220	5.59
.010	.25	.290	7.37
.018	.46	.300	7.62
.020	.51	.310	7.87
.030	.76	.320	8.13
.065	1.65	.350	8.89
.070	1.78	.600	15.24
.100	2.54	.750	19.05
.170	4.32		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Case and mounting dimensions.

(C) denotes changes.

FSC 5840

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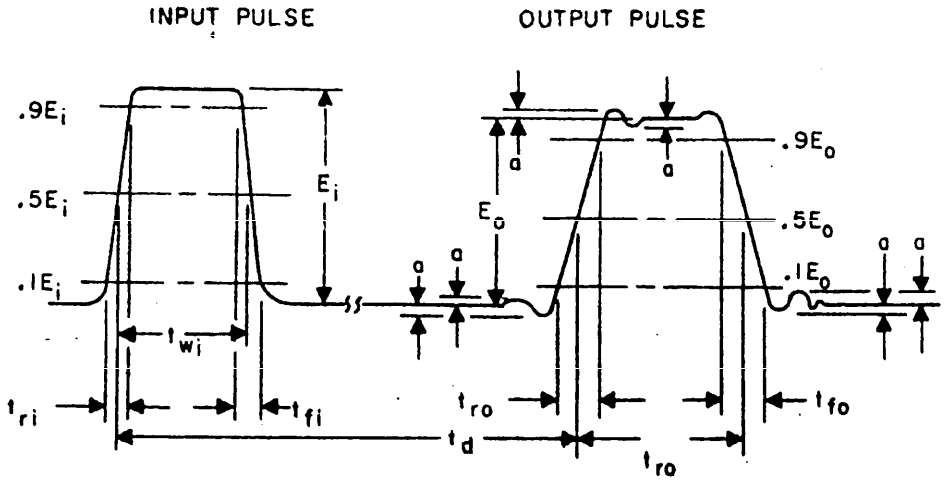
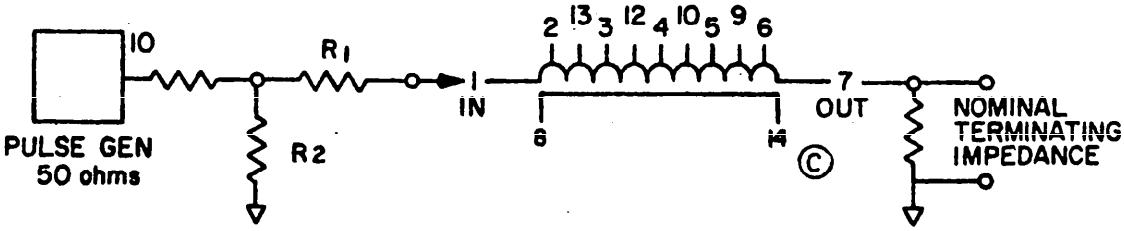


FIGURE 2. Waveform characteristics.



Minimum loss pads

Nominal terminating impedance ±10%	R_1 ±1%	R_2 ±1%
* 50Ω	---	---
100Ω	70.711	70.710
200Ω	173.205	57.735
360Ω	324.037	54.006
500Ω	474.342	52.705

*No pad used when terminating impedance is 50Ω.

FIGURE 3. Test circuit and minimum loss pads for nominal terminating impedances as indicated above.

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ENGINEERING DATA:

Case material: Molded epoxy.

© Operating temperature: -55° to +125°C.

© Storage temperature: -65° to +130°C.

© Weight: 2.5 grams maximum.

Terminals: Pin type.
Terminal material: Copper-nickel, tin plated per MIL-T-10727, type I.

Pulse (see figures 2 and 3).
Input:
Peak pulse voltage: 5.0 volts ±5 percent.

Temperature coefficient of delay: 100 PPM/°C maximum.

Nominal terminating impedance: See table I.

Style: Not applicable.

Figure of merit: Not applicable.

REQUIREMENTS:

Design and construction: See figure 1.

Enclosure: See figure 1.

Capacitors: Chip capacitors shall be items that have been tested and supplied in accordance with MIL-C-55681, when possible.

Total delay time: See table I.

Rise time: See table I.

Attenuation: See table I.

Distortion: ±10 percent at maximum delay.

Thermal shock (burn-in): Method 1011, MIL-STD-883, test condition B, 5 minutes at temperature extremes; prior to first article inspection.
Measurement - Electrical continuity.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.

Insulation resistance: Method 302, test condition A, MIL-STD-202.
100 megohms minimum.

Thermal-age stability: Not applicable.

Sealing: As specified in MIL-D-23859.

Terminal strength: Method 2004, MIL-STD-883, test condition B-2.

Pressure: Not applicable.

Vibration: Method 204, test condition B, MIL-STD-202.
Measurement - Not applicable.

Shock: Method 2002, MIL-STD-883, condition A, 0.5 milliseconds, 5 pulses in X, Y, and Z.

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Immersion: Method 104, test condition A, MIL-STD-202.
Measurement - Insulation resistance.

Moisture resistance: Method 106 of MIL-STD-202.

Solderability: Method 2003, MIL-STD-883, 260°C; 5 s. All terminals shall be tested.

Resistance to solvents: Method 215 of MIL-STD-202.

Flammability: Not applicable.

Life: Not applicable.

Part number: See table I.(Example M23859/23-001)

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TABLE I. Part numbers and electrical characteristics.

Part number M23859/23-					1/ Total delay time	Attenuation (max) per impedance					Rise time (max) at max delay
Nominal terminating impedance (±10%)											
50Ω	100Ω	200Ω	360Ω	500Ω		50Ω	100Ω	200Ω	360Ω	500Ω	
					ns	percent					ns
001	101	201	301	401	5 ±2 ns	3	5	5	5	7	2
002	102	202	302	402	10						3
003	103	203	303	403	15						4
004	104	204	304	404	20						5.5
005	105	205	305	405	25						6
006	106	206	306	406	30						6.5
007	107	207	307	407	35						7
008	108	208	308	408	40 ±2 ns						8
009	109	209	309	409	45 ±5%						9
010	110	210	310	410	50						10
011	111	211	311	411	55						11
012	112	212	312	412	60						12
013	113	213	313	413	65						13
014	114	214	314	414	70						14
015	115	215	315	415	75						15
016	116	216	316	416	80						16
017	117	217	317	417	85						17
018	118	218	318	418	90						18
019	119	219	319	419	95						19
020	120	220	320	420	100			5	5	7	20
021	121	221	321	421	110			6	7	8	22
022	122	222	322	422	120						24
023	123	223	323	423	130						26
024	124	224	324	424	140						28
025	125	225	325	425	150						30
026	126	226	326	426	160						32
027	127	227	327	427	170						34
028	128	228	328	428	180						36
029	129	229	329	429	190						38
030	130	230	330	430	200	3					40
	131	231	331	431	210						45
	132	232	332	432	220						45
	133	233	333	433	230						50
	134	234	334	434	240						50
	135	235	335	435	250						50
	136	236	336	436	260						55
	137	237	337	437	270						55
	138	238	338	438	280						60
	139	239	339	439	290						60
	140	240	340	440	300		5	6	7	8	60
	141	241	341	441	310		7	8	10	10	65
	142	242	342	442	320						65
	143	243	343	443	330						70
	144	244	344	444	340						70
	145	245	345	445	350						70
	146	246	346	446	360						75
	147	247	347	447	370						75
	148	248	348	448	380						80
	149	249	349	449	390						80
	150	250	350	450	400 ±5%		7	8	10	10	80

Not applicable

See footnote at end of table.

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TABLE I. Part numbers and electrical characteristics. - Continued

Part number M23859/23-					1/ Total delay time	Attenuation (max) per impedance					Rise time (max) at max delay
Nominal terminating impedance (±10%)											
50Ω	100Ω	200Ω	360Ω	500Ω		50Ω	100Ω	200Ω	360Ω	500Ω	
					ns	percent					ns
Not applicable	151	251	351	451	410 ±5%		8	10	10	10	85
	152	252	352	452	420						85
	153	253	353	453	430						90
	154	254	354	454	440						90
	155	255	355	455	450						90
	156	256	356	456	460						95
	157	257	357	457	470						95
	158	258	358	458	480						100
	159	259	359	459	490						100
	160	260	360	460	500 ↓		8	10	↓	↓	100
	161	261	361	461	350 ±5%		7	8	10	10	50

1/ Delay between successive taps shall be 1/10 of the total delay time.

Custodians:
Army - EL
Navy - EC
Air Force - 99

Review activities:
Army - AR
Navy - AS, OS, SH
Air Force - 17

User activities:
Army - MI
Navy - MC

Preparing activity:
Air Force - 11

Agent:
DLA - ES

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